

System Requirements Specification: Memento

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1. Description of the Subject Area

The subject area is location-based social memory sharing. Momento is a mobile application that allows users to drop digital time capsules — called "momentos" — at specific GPS coordinates. These capsules can contain photos, voice notes, or text messages. Other users can only discover and open a capsule when they physically visit that exact location.

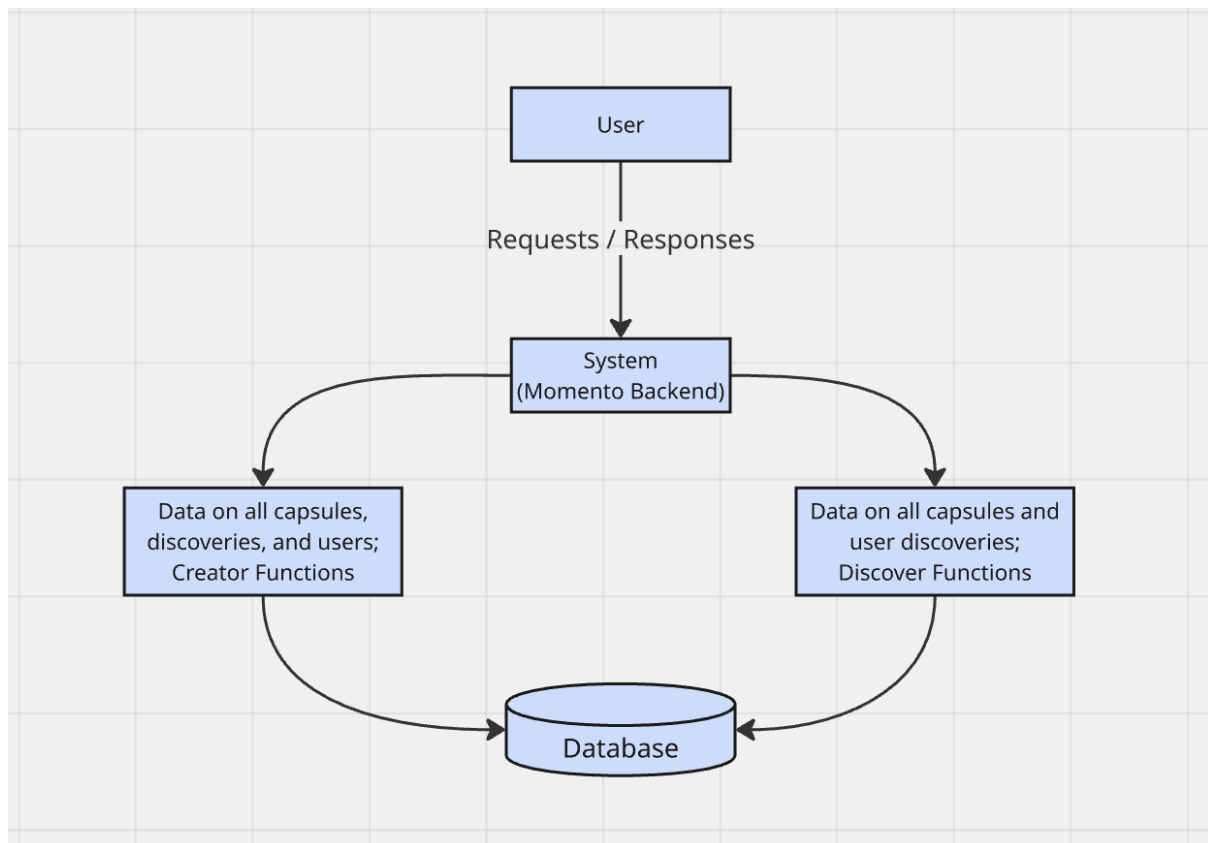
The system allows two categories of users — Creators and Discoverers — though both roles can be played by any registered user. Creators attach digital content to real-world places; Discoverers find and unlock that content by being physically present.

Capsules expire 30 days after being dropped unless discovered, at which point their lifetime is extended by another 30 days. Users earn points for both dropping and discovering capsules, encouraging active participation.

The application is mobile-first, requires user authentication, uses GPS location services, and stores multimedia content in the cloud.

2. Simplified System Model

Figure 1 below illustrates the high-level model of the Momento system.\



[Figure 1: Simplified system model diagram]

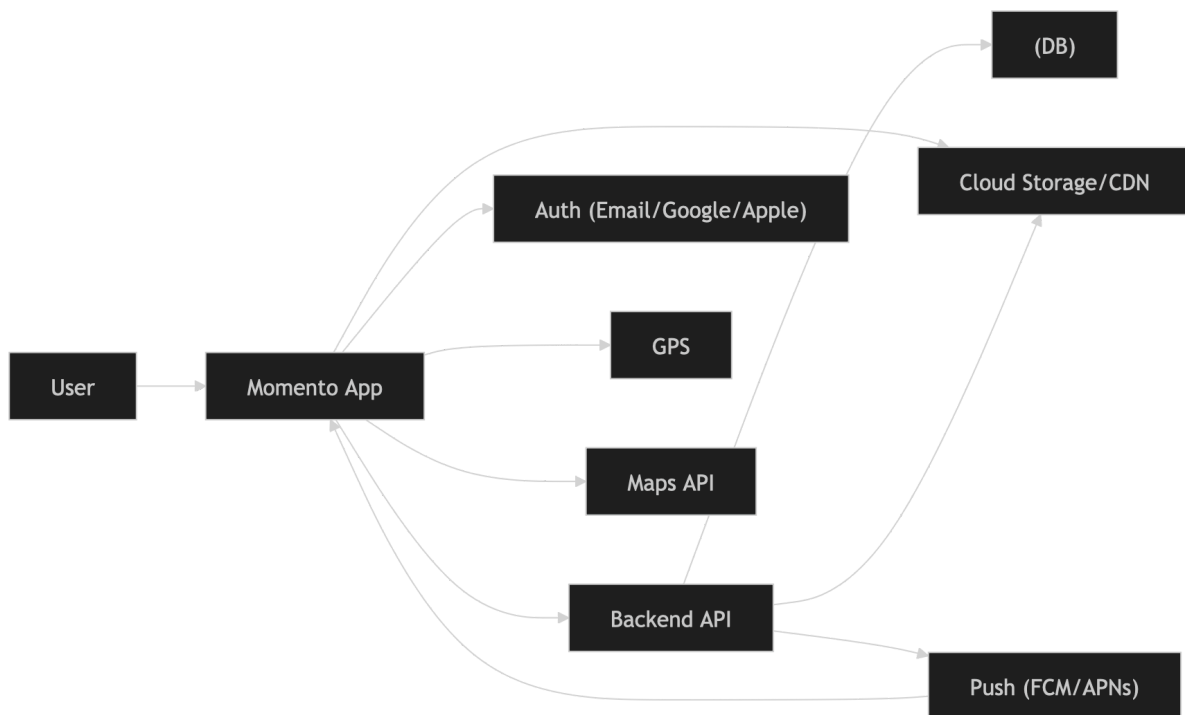
3. Description of the System Model

The interaction flow of the Momento system proceeds as follows:

1. The user logs in or registers via email, Google, or Apple authentication.
2. The system grants access and establishes the user session.
3. The user's GPS location is retrieved and monitored continuously.
4. The system queries the database for capsules near the user's current location.
5. Nearby undiscovered capsules are displayed on the user's map (as markers, without revealing content).
6. When the user physically arrives at a capsule location, the system unlocks the capsule and reveals its content.
7. The system updates the capsule's expiry timer, awards points, and stores the discovery event in the database.
8. The user receives updated information on their profile and history.

4. System Environment Diagram

Figure 2 shows how the Momento system interacts with its external components.



5. Description of the System Environment Diagram

User interaction occurs through the mobile application interface. After authentication, the user gains access to the map view, capsule creation tools, and their personal profile.

The system interacts with the following external services:

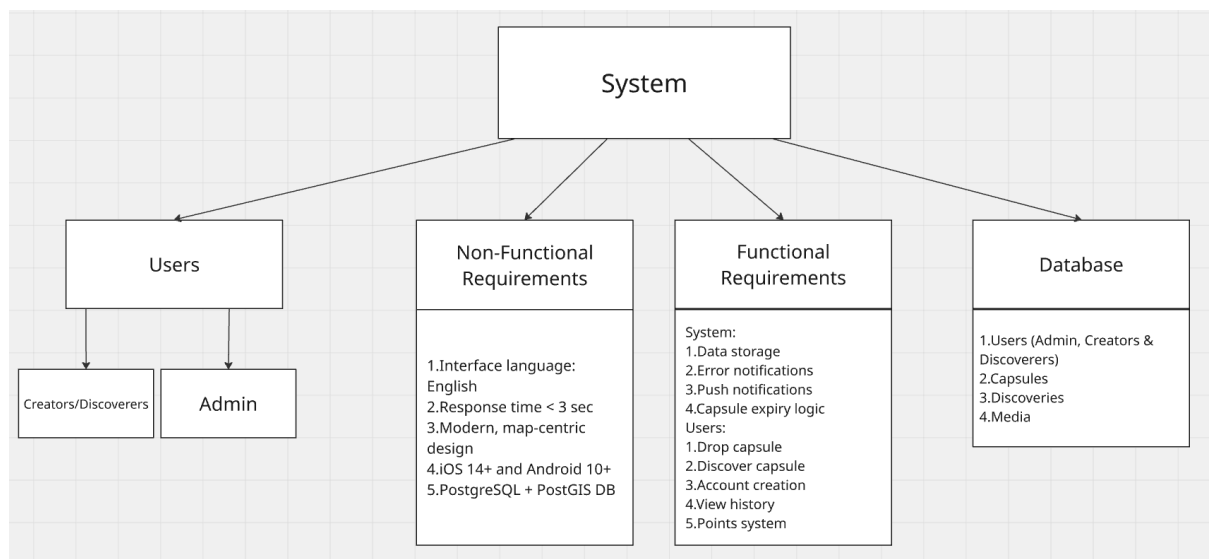
- Google Maps API: provides map rendering, geocoding, and geofencing capabilities.
- Firebase Authentication: handles user identity verification (email/password, Google, Apple login).

- AWS S3 / Firebase Storage: stores all uploaded media files (photos, voice notes).
- Firebase Cloud Messaging (FCM): delivers push notifications for capsule discovery events and expiry warnings.

The backend (Java + Spring Boot) receives user requests, processes business logic, communicates with the PostgreSQL/PostGIS database for capsule and user data, and returns responses to the mobile client.

6. Hierarchical View Diagram

Figure 3 shows the hierarchical breakdown of the Momento system into its main opinion areas.



[Figure 3: Hierarchical view diagram]

The subject area was explored. The information obtained can be divided into opinions covering the following areas:

- Opinions about users — Creators and Discoverers — who interact with capsules differently;
- Opinions about non-functional requirements (performance, platform support, privacy);
- Opinions about functional requirements: system functions and user-facing features;
- Opinions about the database and its objects (capsules, users, discoveries, media).

7. Tabular Diagrams of Opinions

7.1. System Opinion

Source	Input	Process	Output	Place of Assignment
User	Registration info: • username • password • email	User registration in database	• Account created • Error message	Screen, database
User	Authorization info: • username • password	User authorization	• Session token granted • Error message	Screen
User	Capsule data: • media content • GPS coords • timestamp	Capsule creation and storage	• Capsule saved • Marker on map • Error message	Screen, database, storage
System	Scheduled timer check	Capsule expiry validation	• Expired capsules deleted • Notifications sent	Database, FCM

7.2. User – Creator Opinion

Source	Input	Process	Output	Place of Assignment
Creator	• Media (photo / voice / text) • GPS location • Optional message	Drop a capsule at current location	• Capsule stored • Map marker appears • Points awarded	Screen, database, storage
Creator	No input	View own dropped capsules	List of own capsules with status and expiry info	Screen
Creator	Capsule ID	Delete own capsule	• Capsule removed from map • Media deleted from storage	Screen, database, storage

7.3. User – Discoverer Opinion

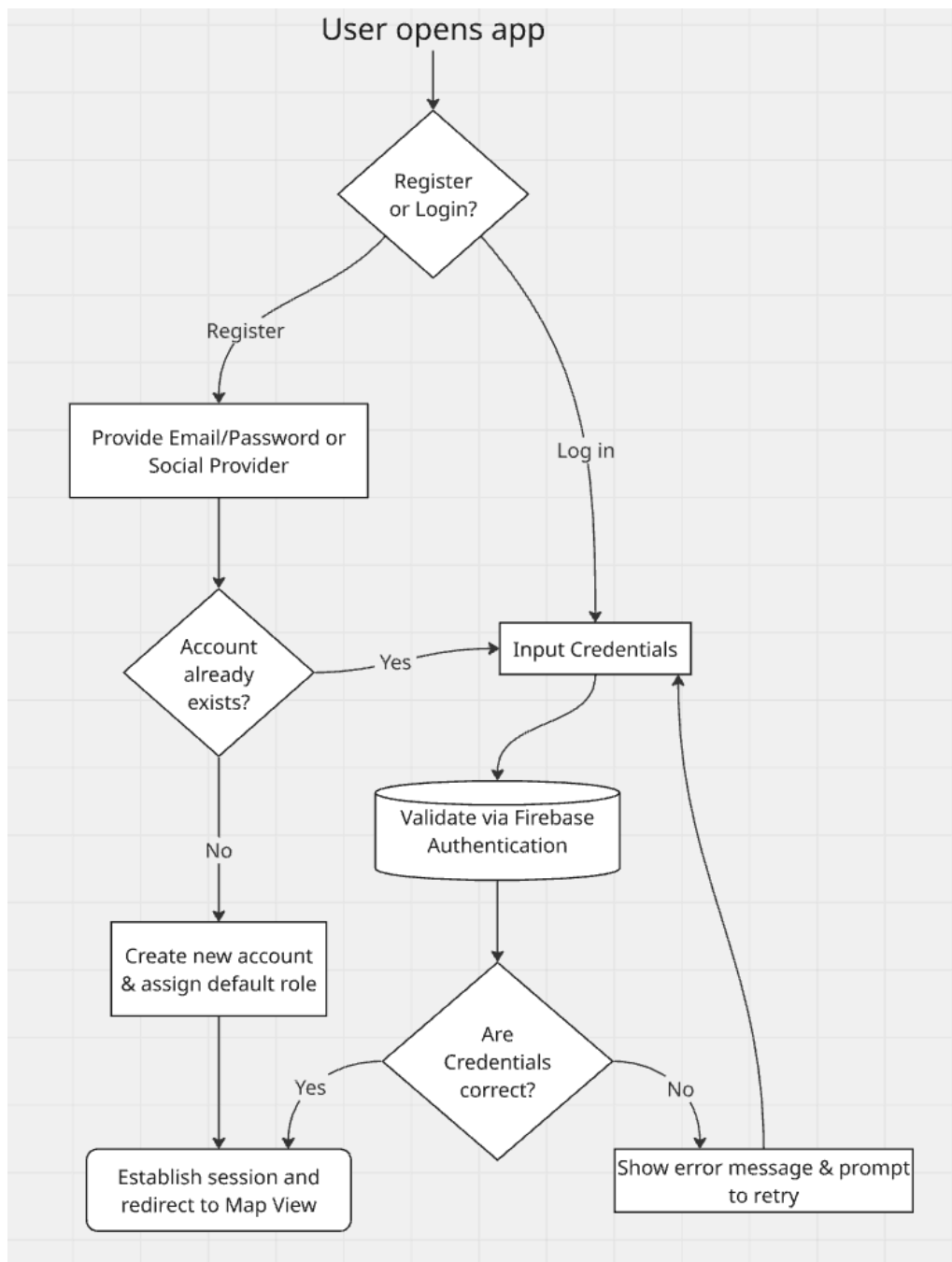
Source	Input	Process	Output	Place of Assignment
Discoverer	Current GPS location	Request nearby capsules	Map markers of undiscovered capsules nearby	Screen
Discoverer	Physical arrival at capsule location	Unlock capsule content	• Capsule content revealed • Expiry extended +30 days	Screen, database, FCM

Source	Input	Process	Output	Place of Assignment
			• Points awarded • Push notification sent to creator	
Discoverer	No input	View discovery history	List of all discovered capsules with details	Screen

8. Activity Diagrams

8.1. User Registration / Authorization Activity Diagram

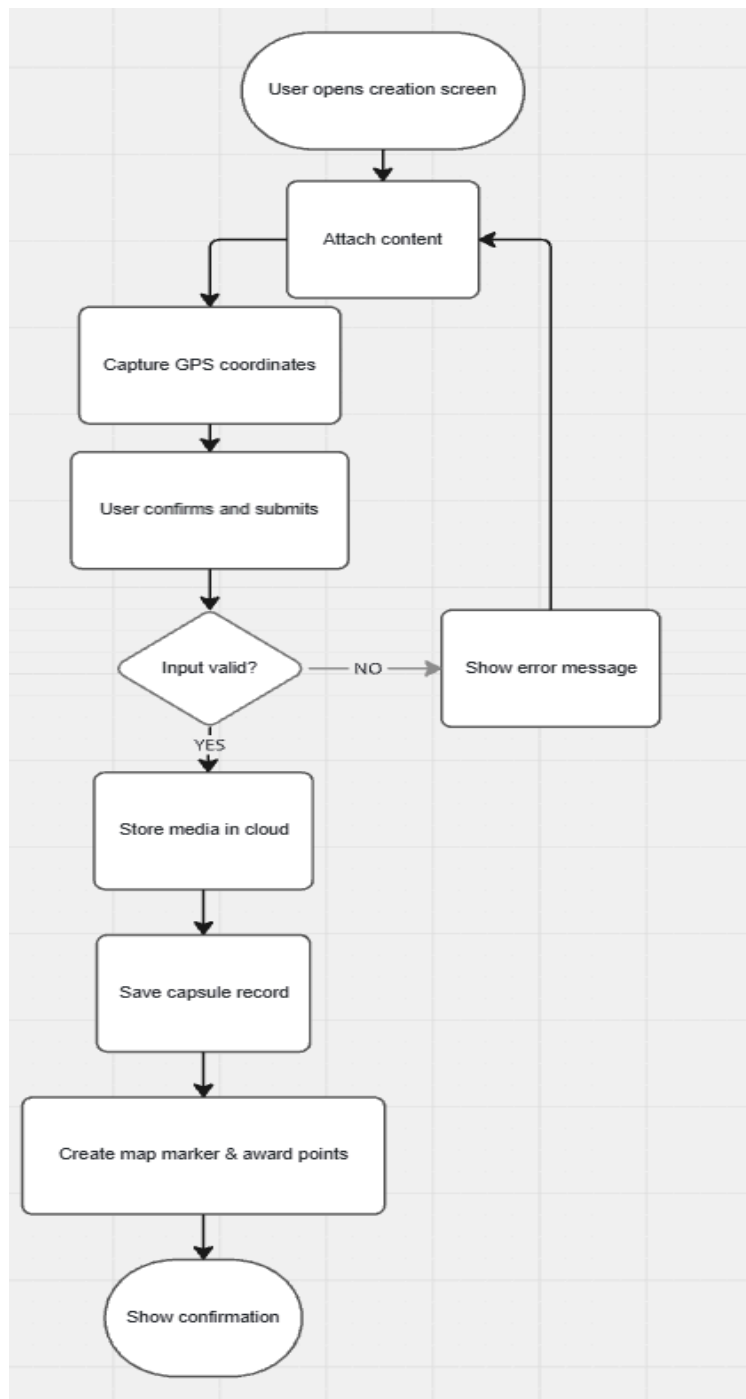
The registration and authorization flow proceeds as follows: the user opens the app and chooses to register or log in. For registration, the user provides an email and password (or selects a social provider). The system checks if the account exists; if not, a new account is created and the user is assigned the default role. For login, the system validates credentials via Firebase Authentication. On success, a session is established and the user is directed to the map view. On failure, an error message is shown and the user is prompted to retry.



[Figure 4: User registration/authorization activity diagram]

8.2. Capsule Drop Activity Diagram

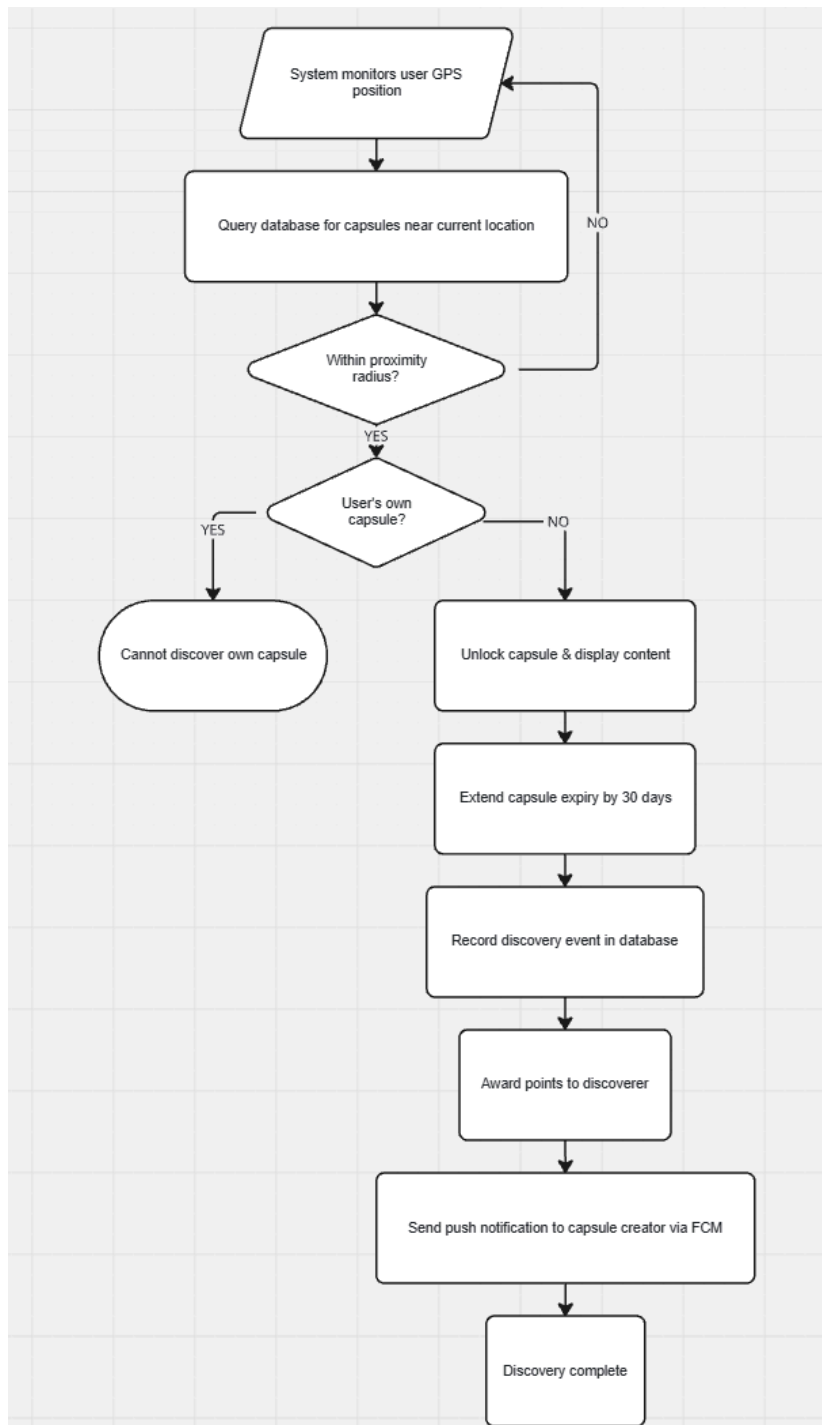
The capsule drop flow proceeds as follows: the user navigates to the creation screen and attaches content (photo, voice note, or text). The app captures the current GPS coordinates. The user confirms and submits. The backend validates the input, stores the media in cloud storage, and saves the capsule record (including location, media URL, creator ID, and expiry timestamp) to the database. A map marker is created, points are awarded, and the user receives a confirmation.



[Figure 5: Capsule drop activity diagram]

8.3. Capsule Discovery Activity Diagram

The discovery flow proceeds as follows: the system continuously monitors the user's GPS position and compares it against capsule locations in the database. When the user enters the proximity radius of a capsule, the system unlocks it and displays the content. The capsule's expiry is extended by 30 days, the discovery event is recorded, the discoverer is awarded points, and the capsule creator receives a push notification via FCM.



[Figure 6: Capsule discovery activity diagram]

9. System Requirements

9.1. Functional Requirements

1. User Classification.

1. User – Creator: any registered user who drops capsules at physical locations.
2. User – Discoverer: any registered user who physically visits locations and unlocks capsules.
3. Note: both roles are available to every registered user simultaneously.

2. User Access to the System.

4. Account creation – a user registers with an email address and password, or authenticates via a supported third-party provider (Google, Apple). The account is stored in Firebase Authentication and a user profile is created in the database.
5. User authorization – the user logs in using registered credentials; the system issues an authenticated session token.
6. Password reset – the user can request a password reset link to their registered email address.

3. Functions Available to a Logged-In User.

7. Drop a capsule – a logged-in user can create a capsule at their current GPS location by attaching a photo, a voice note, or a text message (or a combination). The capsule is stored with a 30-day expiry timer.
8. Map view – the user can view a real-time map showing the markers of undiscovered capsules near their current location. Capsule content is not revealed until the user is physically present.
9. Discover a capsule – when the user physically arrives within the proximity radius of a capsule location, the system automatically unlocks the capsule and displays its content.
10. Capsule expiry extension – upon discovery, the capsule's expiry timer is extended by 30 days. If no discovery occurs within 30 days of the last extension, the capsule is permanently deleted.
11. View dropped capsules – the creator can view a list of all capsules they have dropped, including status (active/expired) and remaining time.
12. View discovery history – the discoverer can view a list of all capsules they have discovered, with date and content.
13. Delete own capsule – the creator can delete their own capsule at any time, removing it from the map and deleting its media from storage.
14. Points system – users earn points for dropping capsules and for discovering capsules. Points are displayed on the user profile.
15. User profile – each user has a profile page displaying their username, total points, number of capsules dropped, and number of capsules discovered.
16. Push notifications – the system sends push notifications to the capsule creator when their capsule is discovered, and sends expiry warning notifications 3 days before a capsule expires.

17. A user may not drop more than one capsule at the exact same GPS coordinates within 24 hours.
18. A user cannot discover their own capsule.

9.2. Non-Functional Requirements

1. Mobile Application Requirements.

19. Application interface language: English.
20. Design: clean, modern, map-centric UI consistent with mobile design guidelines (Material Design / Apple HIG).
21. No advertisements in the base version.
22. API response time: less than 3 seconds under normal network conditions.

2. Platform Requirements.

23. The application must support both iOS (version 14+) and Android (version 10+).
24. GPS and location permissions must be requested at runtime.
25. The application must function on devices with screen sizes ranging from 4.7 to 6.7 inches.

3. Security and Privacy Requirements.

26. All API communication must use HTTPS.
27. User passwords must be managed exclusively through Firebase Authentication (no raw password storage).
28. Precise user location data must not be stored in the database beyond what is necessary for capsule proximity matching.

4. Process and Development Requirements.

29. Frontend development framework: React Native with Expo.
30. Backend development framework: Java with Spring Boot.
31. Database: PostgreSQL with PostGIS extension for geospatial queries.
32. Media storage: AWS S3 or Firebase Storage.
33. Authentication provider: Firebase Authentication.
34. Push notifications: Firebase Cloud Messaging (FCM).
35. Map provider: Google Maps API.
36. Build tool: Gradle.